DSA Experiment 7:Evaluation of postfix expression

Name:Rushita Kiran Sontakke **PRN:B25ET1051**
Batch:B1 **Class:SY1**

```
#include <stdio.h>
#include <ctype.h>
#include <stdlib.h>

#define MAX 100

int stack[MAX];
int top = -1;

void push(int x) {
    stack[++top] = x;
}

int pop() {
    return stack[top--];
}

int evaluatePostfix(char* exp) {
    int i;
    for (i = 0; exp[i] != '\0'; i++) {
        char ch = exp[i];


        if (isdigit(ch)) {
            push(ch - '0');
        }

        else {
            int v2 = pop();
            int v1 = pop();

            switch (ch) {
                case '+':
                    push(v1 + v2);
                    break;
                case '-':
                    push(v1 - v2);
                    break;
                case '*':
```

```
                push(v1 * v2);
                break;
            case '/':
                push(v1 / v2);
                break;
        }
    }
  }
  return pop();
}

int main() {
  char exp[100];

  printf("Enter postfix expression: ");
  scanf("%s", exp);

  printf("Result = %d\n", evaluatePostfix(exp));
  return 0;
}
```

```
Output

Enter postfix expression: 765+*
Result = 77
```